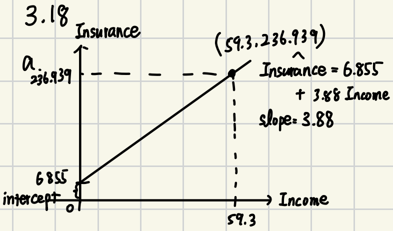
3.18

a 

When the sample of income is 59.3, the sample of insurance is $6.855 + 3.88 \times 59.3 = 236.939$

b.

Point estimate: Since slope of linear regression is 3.88, it means that when income increase \$1000, the insurance will increase \$3880.

95% Interval estimate: $[b_2 \mp t_{\frac{\alpha}{2}}(N-2)\, se(b_2)] = [3.88 \mp t_{0.025}(18)\ 0.112] = [3.88 \mp 2.101 \times 0.112] = [3.644688, 4.115312]$ (in thousands)

It means that in repeated sampling, about 95% intervals constructed in this way will contain the true value of β_2

c. $se(b_1 + 100b_2) = \sqrt{se(b_1)^2 + 10000\, se(b_2)^2 + 2 \cdot 1 \cdot 100\, cov(b_1, b_2)} = \sqrt{7.383^2 + 1.12^2 \times 10^{-2} \times 10^4 + 200 \times (-0.746)} = 5.545$

99% Interval estimate: $[6.855 + 3.88 \times 100 \mp t_{0.005}(18) \times 5.545] = [6.855 + 388 \mp 2.878 \times 5.545] = [378.8965, 410.8135]$

d. $H_0: \beta_2 = 5 \quad H_1: \beta_2 \neq 5$

Test Statistic: $t(N-2) \sim \frac{b_2 - \beta_2}{se(b_2)} = \frac{3.88-5}{0.112} = -10$ Rejection Region: $\{t : |t| > t_{0.025}(18) = 2.101\}$

$\therefore -10 \in$ Rejection Region $\Rightarrow$ Reject H_0 $\Rightarrow$ It has enough evidence to show that $\beta_2 \neq 5$

e. $H_0: \beta_2 = 1 \quad H_1: \beta_2 > 1$

Test Statistic: $t(N-2) \sim \frac{b_2 - 1}{se(b_2)} = \frac{2.88}{0.112} = 25.71$ Rejection Region $= \{t : t > t_{0.01}(18)\} = \{t : t > 2.552\}$

$\because 25.71 \in$ Rejection Region $\Rightarrow$ Reject H_0 $\Rightarrow$ It has enough evidence to show that $\beta_2 > 1$

3.23

(a) Consider the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$. To find the marginal effect, we calculate

$$\frac{d\,PRICE}{d\,SQFT} = 2\alpha_2 SQFT$$

Since original size ($SQFT$) is 20 hundreds feet. We can know that marginal effect is $40\alpha_2$. To test the hypothesis, we have $H_0 : 40\alpha_2 \leq 13$ v.s. $H_1 : 40\alpha_2 > 13$. In other words,

$$H_0 : \alpha_2 \leq \frac{13}{40} \text{ v.s. } H_1 : \alpha_2 > \frac{13}{40}$$

At 5% level of significance, test statistic is

$$t = \frac{\alpha_2 - \frac{13}{40}}{se(\alpha_2)} = -26.72875$$

, and rejection region is

$$\{t\,|t > t_{0.05}(498) = 1.647919\}$$

. And the p-value is close to 1, which is greater than 0.05.

Hence, the test statistic is NOT in the rejection region. Therefore, we can conclude that we don't have enough evidence to reject H_0, that is, we do not have sufficient evidence to conclude that when $SQFT = 2000$ the marginal effect exceeds $13,000$.

(b) Since $SQFT$ become 40 hundreds feet, the marginal effect become $\frac{d\,PRICE}{d\,SQFT} = 2\alpha_2 SQFT = 80\alpha_2$. To test the hypothesis, we have

$$H_0 : \alpha_2 \leq \frac{13}{80} \text{ v.s. } H_1 : \alpha_2 > \frac{13}{80}$$

At 5% level of significance, test statistic is

$$t = \frac{b_2 - \frac{13}{80}}{se(b_2)} = 4.189467$$

, and rejection region is $\{t\,|t > t_{0.05}(498) = 1.647919\}$. And the p-value is 1.65395×10^{-5}, which is smaller than 0.05.

Hence, the test statistic is in the rejection region. Therefore, we can conclude that we have enough evidence to reject H_0, that is, we have sufficient evidence to conclude that when $SQFT = 4000$ the marginal effect exceeds $13,000$.

(c) By running the code, we have the linear regression $Price = 93.56585 + 0.184519 SQFT^2 + e_i$. The expected price is

$$E[PRICE \mid SQFT = 20] = 93.56585 + 0.184519 * 20^2 = 167.37345$$

The 95% confidence interval of expected price is

$$[E[PRICE|SQFT] - t_{0.025}(498) \times se(\alpha_1 + 20^2\alpha_2), E[PRICE|SQFT] + t_{0.025}(498) \times se(\alpha_1 + 20^2\alpha_2)]$$

, which is $[158.0481, 176.6988]$. It means that for repeated sampling, about 95% of confidence intervals constructed this way will contain the true expected value.

(d) The sample mean price of houses with $SQFT = 20$ is 163.3333. Since this value lies within the 95% confidence interval obtained in part (c), the sample average is compatible with the regression result.

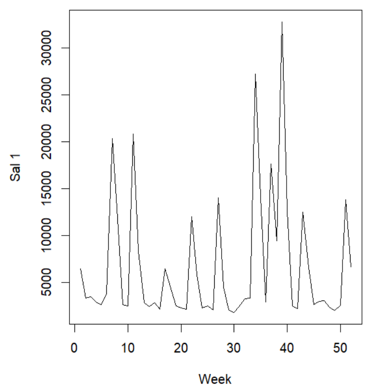
3.31

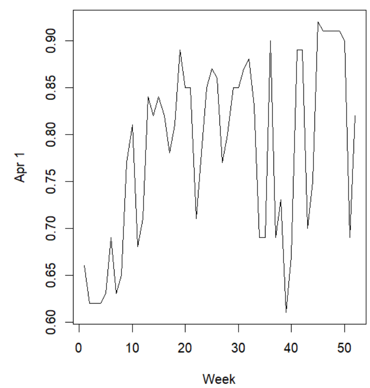
(a) The figure below is summary statistics of SAL1 and APR1.

```
Summary Statistics of SAL1:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1762    2485    3136    6719    8612   32820
Summary Statistics of APR1:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.6100  0.6900  0.8100  0.7825  0.8625  0.9200
```

The standard devistion of SAL1 is 6915.083, and standard devistion of APR1 is 0.09803711.

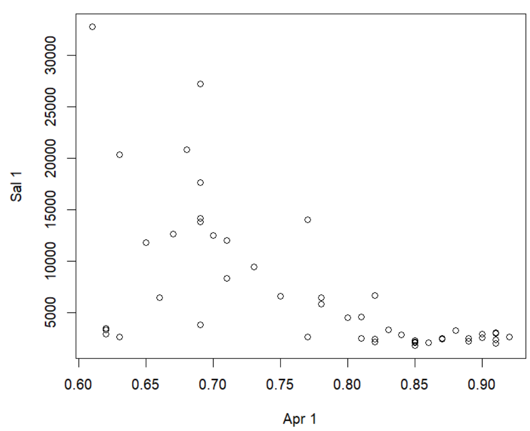
We plot the line chart of SAL1 v.s. Week and APR1 v.s. Week.





Sales fluctuate considerably from week to week, with several large spikes in the time series. In contrast, prices vary within a relatively narrow range and change more smoothly over time.

(b) The figure below is a scatter plot of SAL1 v.s. APR1.



The scatter plot suggests an inverse relationship between SAL1 and APR1. Sales tend to be higher when the price is lower, which is consistent with economic intuition: lower prices attract more consumers to purchase the product.

(c) According to the data, we have the following linear regression:

$$SAL1 = 40714.22 - 434.4473PRICE1 + e_i$$

It means that the point estimate for the effect of a one cent increase in APR1 on SAL1 is β_2, which is -434.4473 cans on average.
The 95% confidence interval of β_2 is

$$[b_2 - t_{0.025}(50) \times se(b_2),\ b_2 + t_{0.025}(50) \times se(b_2)]$$

Hence, the confidence interval is $[-592.2897, -276.6049]$.

(d) When the price is 70 cents, the expected value is

$$E[SAL1 \mid PRICE1 = 70] = 40714.22 + -434.4473 \times 70 = 10302.9$$

The 90% confidence interval of expected sales is

$$[E[SAL1 \mid PRICE1] - t_{0.05}(50) \times se(\beta_1 + 70\beta_2),\ E[SAL1 \mid PRICE1] + t_{0.05}(50) \times se(\beta_1 + 70\beta_2)]$$

, which is $[8624.934, 11980.87]$.

(e) The elasticity is defined as

$$\epsilon = \frac{d\,SAL1}{d\,PRICE1} \cdot \frac{PRICE1}{SAL1} = \beta_2 \cdot \frac{78.25}{6718.712}$$

, where 78.25 is the sample mean of price (in cents), and 6718.712 is the sample mean of unit sales.

The 95% confidence interval is

$$[\frac{78.25}{6718.712}b_2 - se(\frac{78.25}{6718.712}b_2) \times t_{0.025}(50),\ \frac{78.25}{6718.712}b_2 + se(\frac{78.25}{6718.712}b_2) \times t_{0.025}(50)]$$

, which is $[-6.898149, -3.221501]$.

Because the absolute value of elasticity is greater than 1 for the entire interval, demand for brand no.1 tuna appears to be price elastic. This implies that a 1% increase in price leads to more than a 1% decrease in sales.

This makes economic sense because consumers can easily switch to other brands when the price increases, brand-level demand tends to be much more price elastic.

(f) We have the following hypothesis test $H_0 : \varepsilon = -3$ v.s. $H_1 : \varepsilon \neq -3$. From (e), it is equivalent to

$$H_0 : \beta_2 = -3 \cdot \frac{6718.712}{78.25} \text{ v.s. } H_1 : \beta_2 \neq -3 \cdot \frac{6718.712}{78.25}$$

The rejection region is $\{t : |t| > 1.675905\}$. The test statistic is

$$t = \frac{b_2 - (-3) \cdot \frac{6718.712}{78.25}}{se(b_2)} = -2.250572$$

, which is in the critical region. And the p-value is 0.02884204, which is smaller than 0.1.

Therefore, at the 10% significance level, we can conclude that we have enough evidence to reject the null hypothesis that the price elasticity equals -3.